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**Gateway API: what is it
and how does it fit
for LLM self-hosting?**



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Dataroots



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Dataroots



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- Part 1: Gateway API? What is it? (Alaeddine)
- Part 2: Self-hosting LLMs with Gateway API (Alexander)

Hopefully not you guys





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The **problem** with Ingress?

1

Limited Role Model

No distinction between infra team (load balancers) and app team (routing)

2

HTTP-only Focus

No native support for TCP, UDP, gRPC or advanced traffic policies

3

Spec Gaps

Features like traffic splitting, header modification require non-standard workarounds

4

Annotation Overload

Controllers use proprietary annotations for TLS, auth, rate limiting — no standard spec

5

Vendor lock-in

Migrating between ingress controllers requires rewriting all configuration



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What is the **Gateway API**?

The Gateway API is a collection of Kubernetes CRDs (Custom Resource Definitions) that provide a standard, expressive, and extensible API for traffic routing — maintained by the SIG Network working group.



GA since Kubernetes 1.28

Graduated to stable — production ready



Role-oriented design

Cluster operators, app developers, and infra providers have separate concerns



Protocol breadth

HTTP, HTTPS, TLS, TCP, UDP, gRPC natively supported



Portable config

Standardized spec — no proprietary annotations needed



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Core components



GatewayClass

- Defines the type of the loadbalancer
- Created and managed by infrastructure providers
- Cluster scoped
- References a controller implementation (nginx, istio, envoy...)

Gateway

- Instantiates a load balancer from a GatewayClass
- Configures listeners: ports, protocols, TLS certificates
- Namespace scoped
- Can allow/deny routes from specific namespaces

Route

Supported route types

HTTP / HTTPS

TCP

TLS

UDP

Grpc



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Ingress vs Gateway API

✗ Ingress

✓ Gateway API

Spec standardization

✗ Annotations per controller

✓ Standard CRDs — no annotations

Role separation

✗ Single resource, all roles

✓ GatewayClass / Gateway / Route

Protocol support

✗ HTTP/HTTPS only

✓ HTTP, gRPC, TLS, TCP, UDP

Traffic splitting

✗ Non-standard / not supported

✓ Native HTTPRoute weight rules

Header manipulation

✗ Annotation-based, not portable

✓ Native request/response filters

Multi-namespace routing

✗ Limited / manual workaround

✓ Built-in cross-namespace support

Extensibility

✗ Custom annotations (risky)

✓ Policy Attachment extension model



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Follow the demo GitHub project





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Demo? What can go wrong



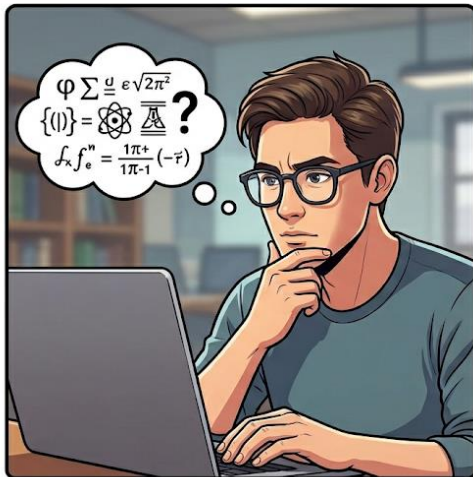


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Self-hosting LLM's: Gateway Inference Extension

USER PROMPT:



LLM RESPONSE:





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- What do you need to self-host an LLM?
- Which exactly issues does the Gateway Inference Extension address?
- Practical solution



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Typical LLM APIs

bash

```
curl https://api.openai.com/v1/chat/completions \  
-H "Content-Type: application/json" \  
-H "Authorization: Bearer $OPENAI_API_KEY" \  
-d '{  
  "model": "gpt-4o-mini",  
  "messages": [  
    {"role": "system", "content": "You are a concise assistant."},  
    {"role": "user", "content": "What is 2+2?"}  
  ],  
  "max_tokens": 64,  
  "temperature": 0.7  
'
```

bash

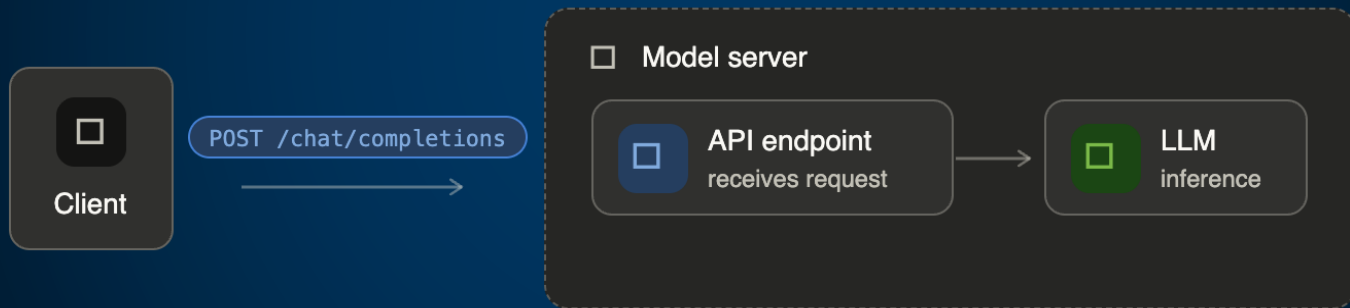
```
curl https://api.openai.com/v1/embeddings \  
-H "Content-Type: application/json" \  
-H "Authorization: Bearer $OPENAI_API_KEY" \  
-d '{  
  "model": "text-embedding-3-small",  
  "input": "The Mediterranean is a sea connected to the Atlantic."  
'
```



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Self-hosting: model servers

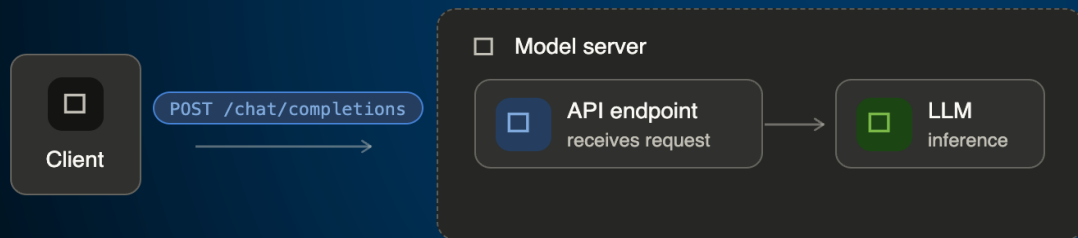




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Self-hosting: model servers



TGI



SGLang



TensorRT-LLM

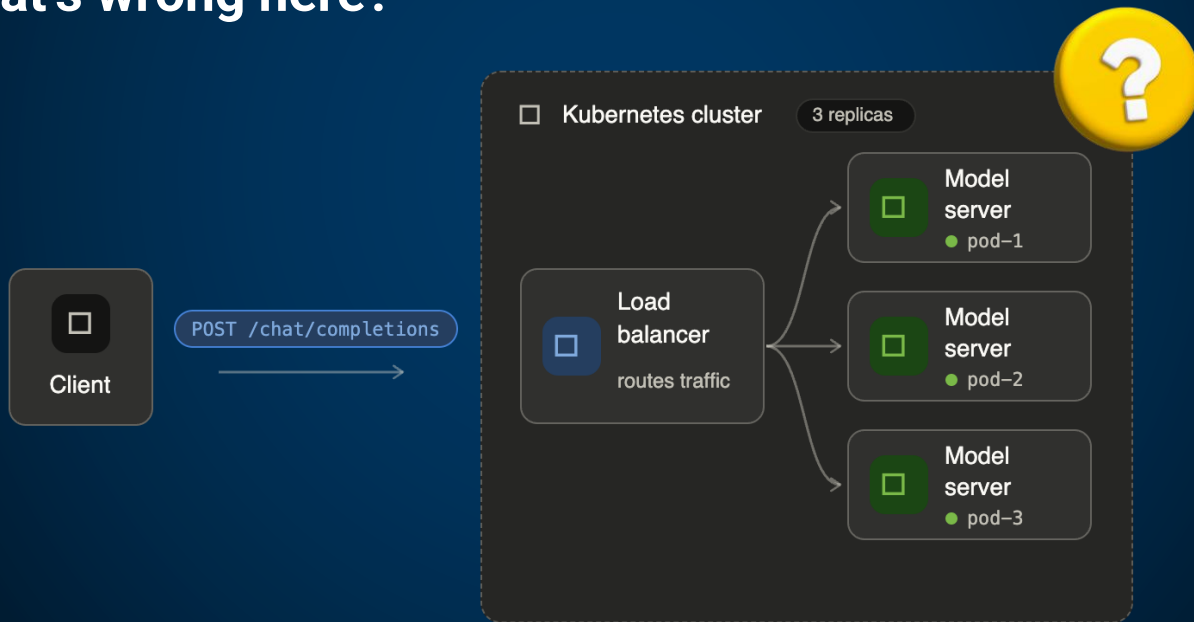




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So... what's wrong here?

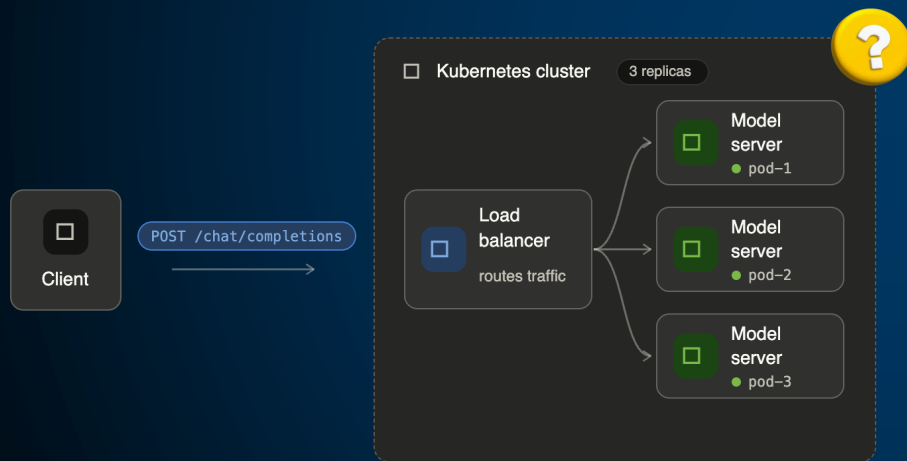




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So... what's wrong here?



Blind to inference state

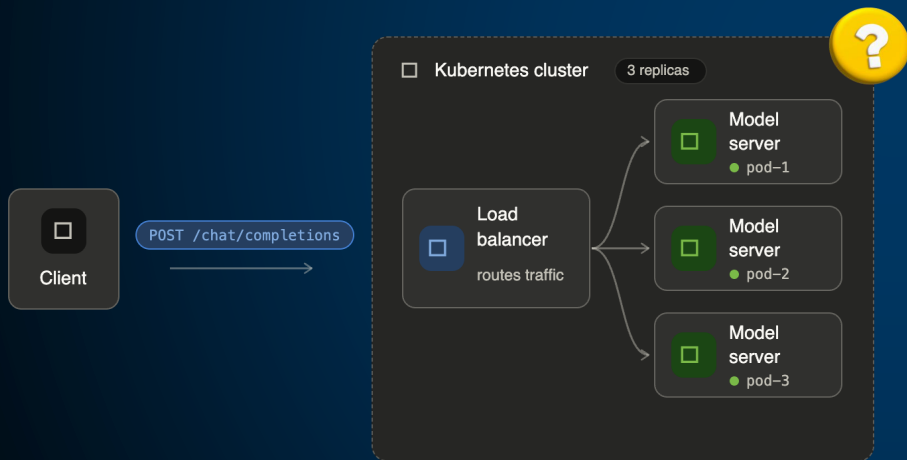
A pod that looks "free" by connection count may be saturated on GPU memory or have a deep request queue



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So... what's wrong here?



Blind to inference state

A pod that looks "free" by connection count may be saturated on GPU memory or have a deep request queue

Can't tell requests apart

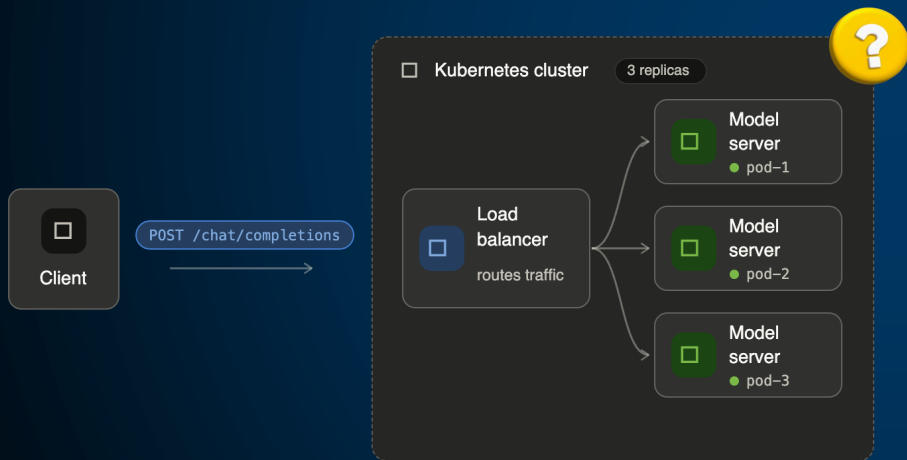
No model identity or priority:
an interactive chat and a batch job compete equally



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So... what's wrong here?



Blind to inference state

A pod that looks "free" by connection count may be saturated on GPU memory or have a deep request queue

Can't tell requests apart

No model identity or priority: an interactive chat and a batch job compete equally

Ignores session state

Stateless routing sends the follow-up to a random pod; no cache → expensive recomputation



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Standard gateways route on connection count



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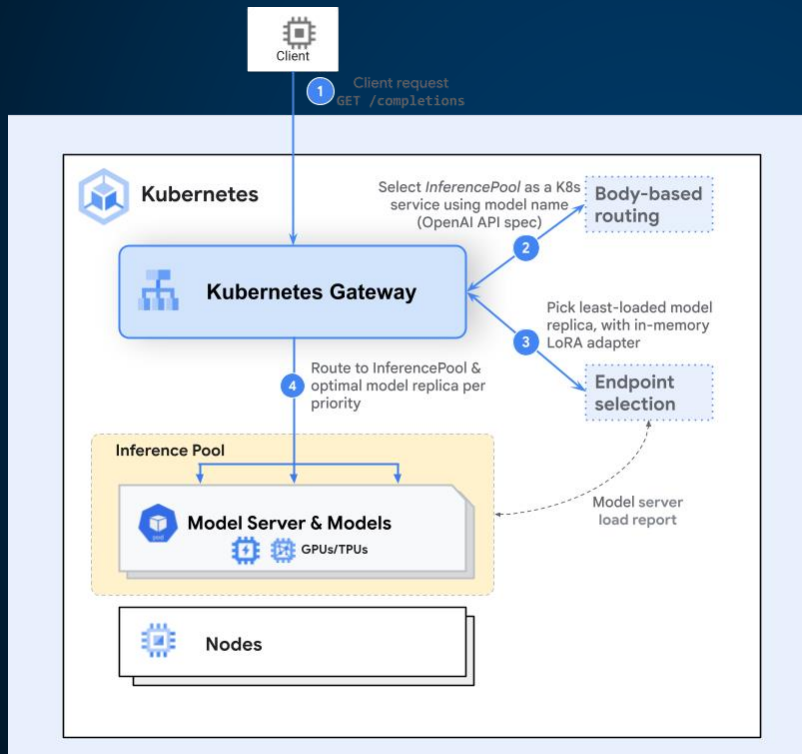
Standard gateways route on connection count

Inference needs routing on GPU state, request priority, and cache locality



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Solution:
Gateway Inference
Extension

GA since Sep, 2025



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Standard Load Balancing **vs** Gateway Inference Extension

LB

✓ GIE

Actual endpoint

Pod

Model Server

Backend abstraction

Service

InferencePool

Endpoint selection mechanism

Internal Kubernetes logic

Endpoint Picker

Endpoint selection metrics

Connections, latency, health, ...

Queue depth, GPU usage, cache state, ...



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